

INTELLIGENT MEDICAL-WASTE COMPLIANCE PLATFORM

MediWaste AI

Intelligent Point-of-Disposal Medical-Waste Segregation,
Compliance Verification & Operational Intelligence Platform

From waste detection to disposal intelligence — MediWaste AI turns a medical-waste image into a policy-controlled disposal decision, verifies the chosen route, explains it from cited evidence, and records an auditable event.



REAL MEDBIN SAMPLE INPUTS



Diagrams in this report are vector illustrations, not screenshots.

Vision observes. **Rules decide.** **RAG supports.** **LLM explains.**

Team Dekhte Aschi

Final Technical & Product Report

BrainChild Season 2.0

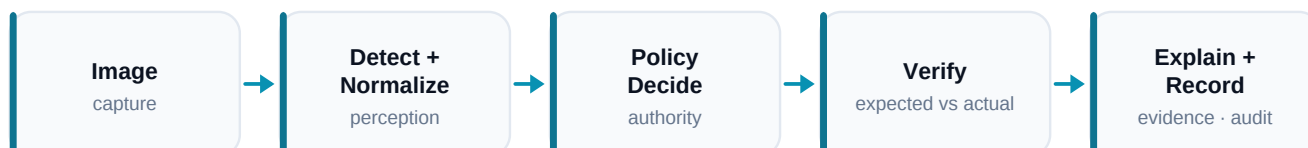
Jagannath University A.I & IT Fest 2026

01 · OVERVIEW

Executive Summary

MediWaste AI transforms medical-waste images into policy-controlled disposal decisions, verifies disposal compliance, provides evidence-grounded explanations, and creates an auditable record of each event.

CORE VALUE CHAIN · CURRENTLY IMPLEMENTED



- **The real problem**

Manual segregation is error-prone, and wrong-stream disposal drives infection and sharps-injury risk, regulatory exposure, and cost. Recognising an object is not the same as making a compliant disposal decision for it.

- **What MediWaste AI does**

It converts a waste image into a deterministic, policy-controlled disposal decision, compares that against the route the operator actually selected, explains the outcome from cited evidence, and writes an auditable event.

- **Current MVP capability**

Image-based analysis (not live video); one deterministic policy pack of seven colour-coded streams; operator-confirmed actual route; best-effort retrieval and explanation with graceful degradation; a SQLite audit trail and an analytics view.

- **Core technical contribution**

A strict decision boundary: a deterministic policy engine — not the language model — is the single authority for waste category and disposal route. Perception, evidence, and explanation are kept separate from the decision.

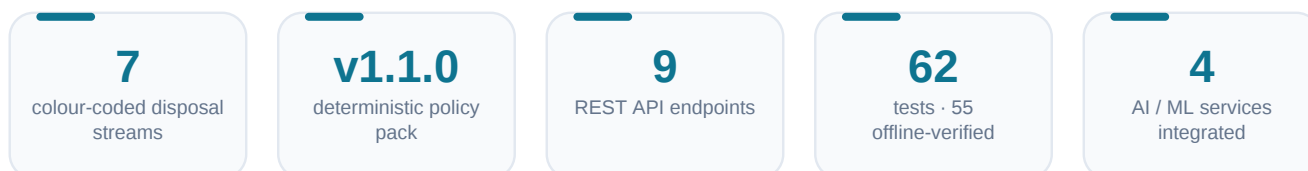
- **AI components**

Roboflow MedBin detector for perception, a CLIP zero-shot visual-context estimate, Pinecone retrieval for evidence, and GPT-OSS-120B (via OpenRouter) for explanation only — all arranged around a team-built policy and compliance core.

- **Intended impact & differentiator**

Designed to reduce segregation errors and make every decision explainable, traceable, and reviewable by ward or station. The perception model is replaceable; the policy, compliance, and audit architecture is the product.

SYSTEM AT A GLANCE · VERIFIABLE FROM THE REPOSITORY



Vision observes. Rules decide. RAG supports. LLM explains.

02 · WHY THIS IS HARD

Problem & Existing Approaches

Medical-waste management is not simply an object-recognition problem. A workable system has to contend with human segregation error, wrong-stream selection, contamination and context, policy ambiguity, and — just as importantly — the absence of evidence and traceability that lets a facility catch repeated violations and act on them.

Human segregation error

Wrong disposal stream

Context / contamination

Policy ambiguity

No evidence

No traceability

Repeated violations

Limited visibility

EXISTING APPROACHES · WHERE THEY FALL SHORT

Existing approach	Core limitation
Manual segregation	Relies on human judgement under time pressure; error-prone and unevenly documented.
Object detector alone	Identifies objects but does not, by itself, determine a policy-compliant disposal route.
Rule-only application	Sound rules, but require a human to identify and enter the object first.
Generic LLM chatbot	Fluent, but not an appropriate deterministic authority for a safety-critical routing decision.
Dashboard-only system	Mostly reactive reporting; no intervention at the point of disposal.

MediWaste AI combines perception, deterministic policy reasoning, verification, evidence retrieval, and auditability into one point-of-disposal workflow.

“Is this just a pretrained model + a website?”

No. The pretrained detector is one replaceable perception component. Everything that makes a disposal decision safe, explainable, and auditable — the canonical waste ontology, the deterministic policy engine, confidence and abstention behaviour, expected-vs-actual verification, the evidence-grounding gate, and the audit trail — is engineering built by the team around that model. Page 5 shows this layer explicitly.

DETECTION ALONE DOES NOT CROSS THIS GAP



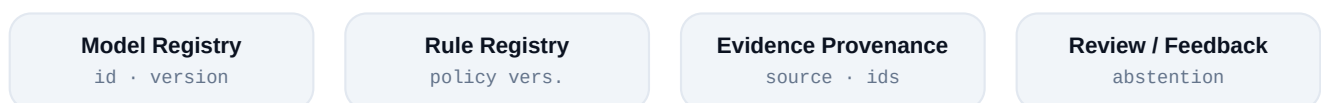
03 · ONE-DIRECTIONAL PIPELINE

System Architecture

Data flows one way, and the decision authority sits in the middle. Perception feeds a deterministic core; evidence and language sit downstream of the decision and can never reach back to change it.



CROSS-CUTTING PROVENANCE & GOVERNANCE



Every stage writes its inputs and outputs to the audit event, so any decision in the pipeline can be reconstructed and reviewed after the fact — the boundary below is enforced structurally, not by convention.

Vision observes. Rules decide. RAG supports. LLM explains.

04 · REUSED VS TEAM-BUILT

AI / ML & Engineering Stack

MediWaste AI integrates four third-party AI/ML services for perception, visual context, retrieval, and language. Each is confined to a specific role — the disposal decision, verification, and audit layers around them are engineered by the team.

INTEGRATED THIRD-PARTY AI / ML SERVICES

Service	Role in the pipeline	Boundary
Roboflow MedBin (medbin_dataset-fqhi7/1)	Object perception and detection	Observes — proposes labels only
CLIP (ViT-B/32)	Zero-shot visual-context estimate	Advisory — never decides
Pinecone (brainchild index)	Evidence retrieval (top-k 8)	Supplies citations only
OpenRouter GPT-OSS-120B	Natural-language explanation	Explains — never decides

WHY THIS IS MORE THAN A PRETRAINED MODEL + A WEB APP

Web UI · Flask REST API (9 endpoints)	DELIVERY
Pretrained perception — Roboflow MedBin detector	REUSED
Canonical normalization · waste ontology	TEAM-BUILT
Deterministic policy engine · v1.1.0 · 7 streams	DECIDES
Confidence & abstention gates	TEAM-BUILT
Expected-vs-actual compliance verification	TEAM-BUILT
Evidence grounding gate → explanation	TEAM-BUILT
SQLite audit trail & analytics	TEAM-BUILT

Exactly one layer is a reused, replaceable perception model. Everything around it — ontology, policy, verification, grounding, and audit — is engineering built by the team. Swap the detector and the compliance architecture still stands.

05 · THE MODEL PROPOSES, THE POLICY DISPOSES

Decision Safety & Responsible AI

Safety-critical routing is never delegated to a probabilistic model. A deterministic, versioned policy engine makes every category-and-route decision, and the system is explicitly allowed to abstain when it is not confident.

DECIDES · DETERMINISTIC

- Deterministic policy engine (policy v1.1.0)
- Expected route from versioned static rules
- Expected-vs-actual compliance verification

SUPPORTS · ADVISORY

- Perception — Roboflow MedBin (observes)
- Visual context — CLIP (estimate)
- Evidence — Pinecone retrieval (supports)
- Explanation — GPT-OSS-120B (describes only)

SAFETY STATES · THE SYSTEM IS ALLOWED TO ABSTAIN

Condition	System behaviour	Signal
Confidence ≥ 0.40, known class	Trusted — policy rule applied	ACCEPT
Confidence below 0.40	Not routed — sent to human review	REVIEW · LOW_CONFIDENCE
Unknown / unmapped class	No route guessed — review	REVIEW · UNKNOWN_CLASS
No expected route	Flagged, not treated as a violation	REVIEW_REQUIRED
No citable evidence	Explanation withheld	SKIPPED_NO_EVIDENCE

Two thresholds are defined in the policy pack: detections at or above 0.40 are routed; those below 0.20 are treated as noise; the band between is sent to review — the system never forces an uncertain prediction into a confident disposal recommendation.

The LLM is not the decision authority. A deterministic, versioned policy engine makes every category-and-route decision; the language model only explains it.

RESPONSIBLE-AI BEHAVIOUR

- **Never defaults to general waste** — unknown and low-confidence cases are routed to review, not silently placed in general waste.
- **Deterministic & versioned** — every event stores the policy version and the rule id that produced the decision.
- **Graceful degradation** — if retrieval or the language model fails, the disposal decision and the audit record still complete.

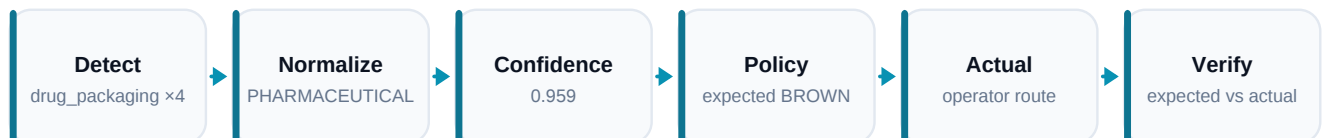
06 · ONE IMAGE, ONE AUDITABLE EVENT

End-to-End Compliance Workflow

The same detection can end in compliance or in a violation depending only on the route the operator selects. Both outcomes below are real pipeline events captured during development — nothing is hardcoded.

**Real MedBin sample input**

Pharmaceutical blister packs
static/samples/sample3.jpg

**CORRECT**

Operator selected BROWN → matches expected.

expected

 **BROWN**

actual

 **BROWN**

event 9a9f2b0f · rule R-PHARMA · v1.1.0

VIOLATION

Operator selected YELLOW → mismatch.

expected

 **BROWN**

actual

 **YELLOW**

event 05b0846e · WRONG_WASTE_STREAM

Both events share the same perception (drug_packaging x4 → PHARMACEUTICAL, 0.959) and the same rule (R-PHARMA); only the operator's selected route differs. Each was retrieved and explained (RAG OK, LLM OK) and written to the audit trail.

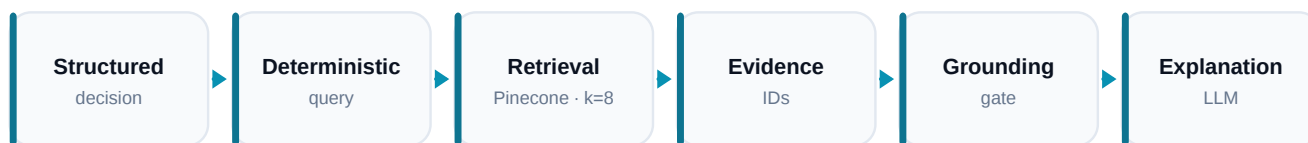
WHAT EACH RUN RECORDS

event id · timestamp · raw detector labels · canonical class · confidence · model id · expected route · rule id · policy version · actual route · compliance status · RAG status · LLM status · evidence IDs.

07 · GROUNDED, OR IT STAYS SILENT

Evidence, Retrieval & Explainability

Explanations are generated only from retrieved, citable evidence. The query is built from the structured decision, not free text, and if nothing relevant is retrieved the language model is not called at all.



No evidence → no grounded explanation. The language model is never invoked without citable evidence IDs; with none, the event records SKIPPED_NO_EVIDENCE and no explanation is fabricated.

- **Deterministic query**

Built from the structured decision — canonical class, disposal stream, and rule — so retrieval targets the policy in force rather than an open-ended prompt.

- **Provenance preserved**

Evidence source and IDs are stored on the event. The pharmaceutical example above carried eight evidence IDs recorded alongside its decision.

- **Retrieval scope**

Retrieval runs against the Pinecone 'brainchild' index with its integrated embedding model, returning up to eight passages per query for the grounding gate to filter.

- **Explanation-only role**

The model describes the policy outcome in natural language; it cannot change the disposal route or override the rule that decided it.

- **Hallucination guard**

Evidence IDs not present in retrieval are stripped, so unsupported citations are never surfaced to the operator.

- **Safe degradation**

Retrieval and LLM calls are isolated; on failure their status is recorded as UNAVAILABLE and the decision still stands.

Index vector count and embedding dimension are inspected at runtime against the live Pinecone index; they are not asserted as fixed figures in this report.

08 · EVERY EVENT RECONSTRUCTABLE

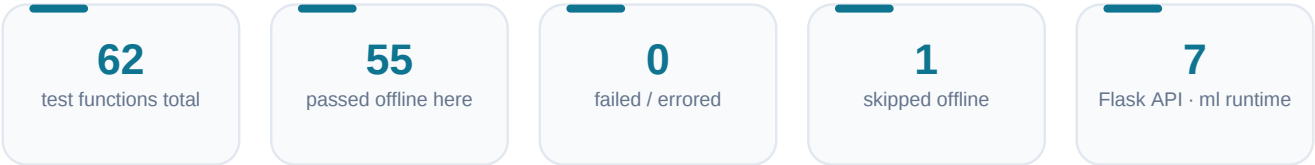
Auditability, Testing & Analytics

Each analysis writes a single, self-describing event. Any past decision can be reconstructed from what the system saw, which rule fired, and the model and policy versions that produced it.

AUDIT EVENT · WHAT EACH ANALYSIS RECORDS

Field group	Captured fields
Identity & time	event id, timestamp
Perception	raw detector labels, canonical class, confidence, model id
Decision	expected route, rule id, policy version
Outcome	actual / selected route, compliance status
Support	RAG status, LLM status, evidence IDs

TESTING · OFFLINE-VERIFIED VS REQUIRES ML RUNTIME



Offline run in this environment: passed 55, failed 0, errored 0, skipped 1. The 7 Flask API tests require the ML runtime and are executed there — deferred, not counted as passed.

ANALYTICS · DEVELOPMENT & TEST DATA (NOT A DEPLOYMENT)



These are real pipeline outputs from development and testing, not hospital operational data. The analytics view aggregates recorded events; richer ward/station attribution is future work.

HOW THE AUDIT TRAIL IS DESIGNED TO SURFACE PATTERNS



09 · TODAY'S MVP VS WHAT'S NEXT

Operational Roadmap

Everything on the left is implemented in the current MVP. Everything on the right is proposed and clearly not yet built — the distinction is kept explicit so no future capability reads as a shipped feature.

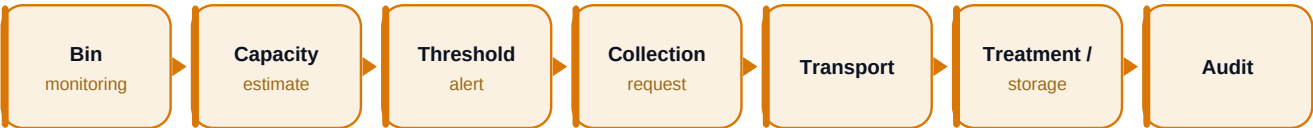
CURRENT MVP · IMPLEMENTED

- Single-image analysis (not live video)
- 7-stream deterministic policy pack (v1.1.0)
- Operator-selected actual route + verification
- Best-effort RAG + LLM explanation (graceful degradation)
- SQLite audit trail + analytics view
- 9-endpoint Flask REST API

PROPOSED · NOT YET IMPLEMENTED

- Live capture / continuous monitoring
- Physical bin recognition & fill-level sensing
- IoT / weight sensors and threshold alerts
- Automated collection routing
- Facility-specific policy profiles
- Larger validation dataset & clinical validation

FUTURE OPERATIONAL LAYER · PROPOSED — NOT IMPLEMENTED



FUTURE EXTENSION · NOT A SHIPPED FEATURE

Bin-capacity and fill-level tracking is a proposed future capability. The current MVP does not sense physical bins or measure fill levels; any such view would be a conceptual prototype, not sensor data.

From point-of-disposal compliance toward facility-wide waste operational intelligence.

10 · FROM ONE STATION OUTWARD

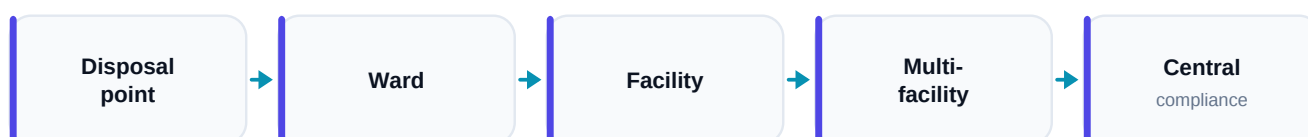
Impact, Scalability & Deployment

The architecture is designed to reduce segregation errors and make every decision traceable and reviewable. Benefits below are stated as intended outcomes of the design, not as measured deployment results.

INTENDED IMPACT · BY DESIGN, NOT MEASURED

- Fewer wrong-stream disposals at the point of decision.
- Traceable, evidence-grounded decisions for every event.
- Safer handling through earlier, consistent segregation.
- Facility-wide compliance visibility over recorded events.

SCALABILITY PATH



The policy engine already carries a facility-profile parameter; today a single 'default' profile is active. Rules and model version are configurable per profile as scale increases.

DEPLOYMENT PROPERTIES

• Replaceable perception

The model id is stamped on every event, so a stronger detector can be swapped in without touching the decision or audit layers.

• Durable audit

A SQLite audit trail today; the same event schema ports to a managed database for multi-facility scale.

• Failure isolation

Retrieval and language calls run best-effort; their failure is recorded but never blocks a disposal decision.

• Event-based & modular

Perception, policy, evidence, and explanation are independent stages joined by a recorded event, not a monolith.

OBSERVED LATENCY · SINGLE SAMPLE, TEAM ML ENV (NOT A BENCHMARK)



Measured once on a single sample and dominated by network round-trips to hosted services. A 4 → 1 CLIP-encode change was verified numerically equivalent; its runtime effect is not yet measured.

11 · HONEST BOUNDARIES

Limitations, Alignment & Conclusion

LIMITATIONS · DISCLOSED

Limitation	Handling / status
Single image, not live video	By design in the MVP; live capture is future work
Perception limited to dataset classes	Detector is replaceable; unknown classes → review
Actual route is operator-selected	Not physically sensed in the MVP
CLIP context is an estimate	Advisory only; not clinical ground truth
RAG / LLM need network + credentials	Degrade gracefully; the decision is unaffected
No clinical / regulatory validation	Not claimed; requires a formal study
Analytics from dev / test data	No operational or deployment claims are made

EVALUATION-CRITERIA ALIGNMENT · OFFICIAL WEIGHTS

Criterion	Addressed by
Innovation (15)	Decision-boundary architecture: the model proposes, the deterministic policy disposes
Problem Solving (20)	End-to-end point-of-disposal workflow with verification, not just detection
Technical Excellence (20)	Policy engine, grounding gate, audit schema, and a 62-test suite
AI Integration (15)	Four AI/ML services with strict, separated roles
User Experience (10)	Operator-confirmed routing, clear compliance signals, analytics view
Scalability (10)	Replaceable perception, facility-profile parameter, portable audit schema
Presentation Skills (10)	This dossier plus an inspectable, reproducible repository

MediWaste AI is not designed merely to recognize medical waste; it is designed to turn recognition into policy-controlled, verifiable, explainable and auditable disposal intelligence.

THE PROGRESSION

What is it? → What does it mean in this context? → Where should it go? → Is the selected route correct? → Why? → What evidence supports it? → What happened? → What patterns are emerging? → How can the facility improve?

From Waste Detection to Disposal Intelligence